

A closed-loop bandage, shown as a digital twin.

SIMULATION ONLY

Patchline demonstrates how a future smart dressing could turn four wound-environment signals into a reviewable action - and stop when a sensor cannot be trusted.

01 SENSE

pH, patch temperature, dressing wetness, wound-fluid glucose

02 LEARN

A small model trains and tests on separate synthetic traces

03 ORCHESTRATE

A quality gate checks the score and may refuse to proceed

04 SHOW

Virtual ultrasound and violet states; evidence-linked handoff

THE LIVE DEMO

From change to restraint

- 1 Select a stable stream. The virtual patch records four generated channels.
- 2 Select a changing stream. The trained model scores a synthetic pattern; the orchestrator explains a virtual response.
- 3 Inject a pH fault. The quality gate cancels virtual outputs and asks for a recheck and human review.

WHAT THE PROTOTYPE PROVES

A visible decision loop

Reproducible sensor streams, manual controls, held-out synthetic evaluation, event history, and a simulated board/firmware view.

Every decision shows its input window and reason. Virtual outputs use abstract levels 0-3; they are not physical settings.

RESEARCH BOUNDARY

No patient data, physical patch, clinical validation, infection diagnosis, or delivered treatment. Synthetic labels test software behavior only. Wound-fluid glucose is not blood glucose. This concept cannot guide care.

EVIDENCE CONTEXT

IWGDF/IDSA: diabetes-related foot infection requires clinical assessment; foot temperature alone is not a diagnostic test. IWGDF wound-healing guidance does not establish this proposed physical-therapy approach as routine care.

Sources: idsociety.org/practice-guideline/diabetic-foot-infections/
onlinelibrary.wiley.com/doi/10.1002/dmrr.3644